جامعة عفت
EFFAT UNIVERSITY
Research
Student Scientific Research Forum & Exhibition

The Effat 7th Undergraduate Research Forum
Saudi Arabia as A Global Hub:
Leveraging International Events
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# AI-DRIVEN MULTI-DISEASE HEALTH RISK PREDICTION SYSTEM

## Research Brief

• Uses three real-world healthcare datasets

• Combines lifestyle factors and clinical indicators

• Applies advanced machine learning models

• Produces clear predictions and recommendations for users

## Introduction

Chronic diseases such as diabetes, sleep disorders, and heart disease are increasing worldwide and often remain undetected until advanced stages.

This project leverages machine learning to analyze lifestyle and clinical data in order to predict health risks early, enabling preventive action and healthier decision-making.

## Methodology and Design

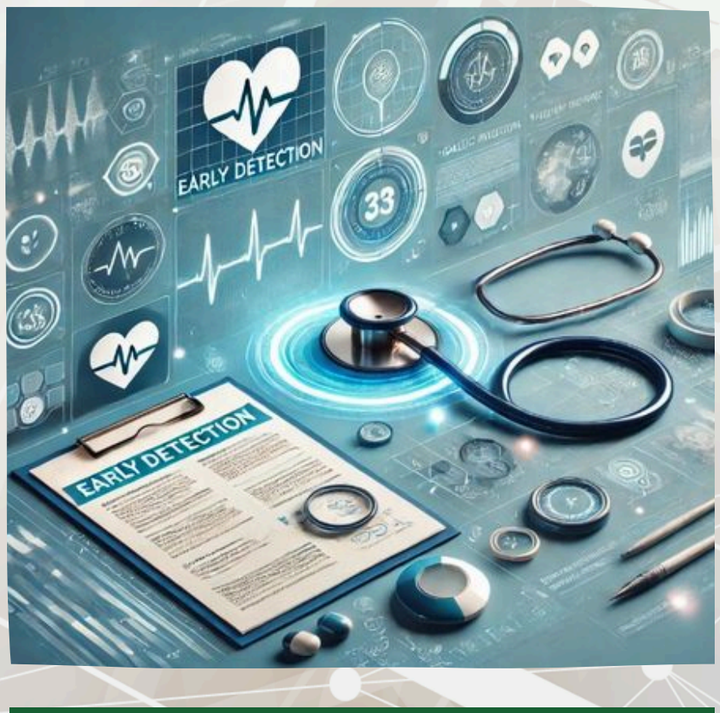
1. **Data Understanding:**Reviewing datasets, features, and class distribution

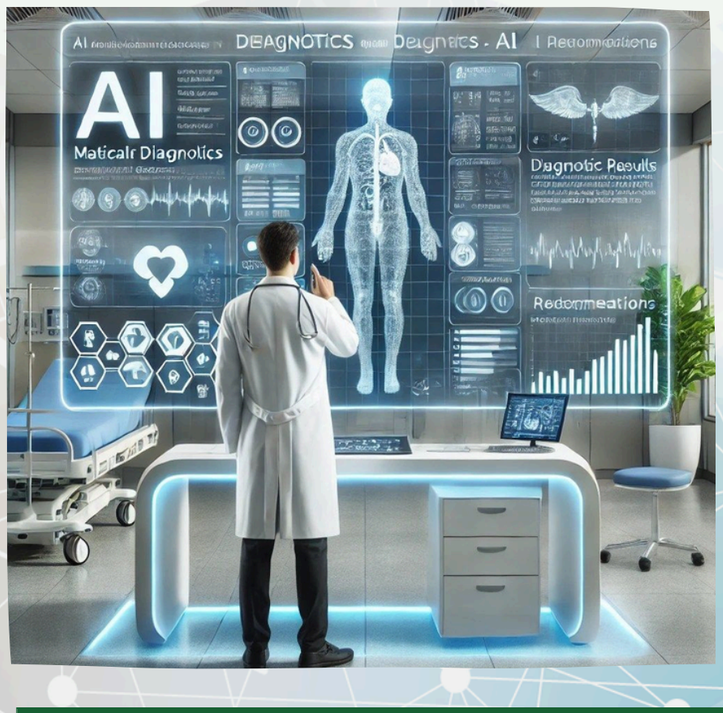
2. **Data Cleaning & Preprocessing:** Handling missing values, encoding, scaling

3. **Exploratory Data Analysis (EDA):**Visualizing patterns and correlations

4. **Model Development:**Testing Logistic Regression, KNN, SVM, Random Forest, XGBoost

5. **Final Prediction System:**User input → Prediction → Risk probability → Recommendations






## Significant Findings

• Combining lifestyle and clinical data improves prediction accuracy

• Random Forest and XGBoost achieved the best performance

• Handling data imbalance using SMOTE significantly enhanced model reliability

• Unified multi-disease prediction offers a holistic health assessment

## Conclusion and Recommendation

- Highlights the effectiveness of AI in preventive healthcare
- Introduces a unified system for early health risk detection
- Supports medical decisions but does not replace diagnosis

Promotes:
1. Healthier lifestyle choices
2. Early medical consultation
3. Increased health awareness

## Group Members

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